

Assignment No.2 (Course STA631)

FALL 2025 (Total Marks 20)

Student ID: BC....

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Deadline: 30 December, 2025

Your Assignment must be uploaded/ submitted before or on **11:59pm 2025**

(STUDENTS ARE **STRICTLY DIRECTED** TO SUBMIT THEIR ASSIGNMENT BEFORE OR BY DUE DATE. NO ASSIGNMENT AFTER DUE DATE WILL BE ACCEPTED VIA E. MAIL).

Rules for Marking

It should be clear that your Assignment will not get any credit IF:

- The Assignment submitted, via email, *after the due date*.
- The Submitted Assignment is not found as an MS Word document file.
- There will be unnecessary, extra or irrelevant material.
- The Statistical notations/symbols are not well-written, i.e., without using **Math Type** software.
- The Assignment will be copied from handouts, internet or from any other student's file. Copied material (from handouts, any book or by any website) will be awarded **ZERO MARKS**. It is **PLAGIARISM** and an **Academic Crime**.
- The medium of the course is **English**. Assignment in **Urdu or Roman** languages will not be accepted.

An assignment means Comprehensive yet precise, accurate details about the given topic, quoting different sources (books/articles/websites etc.). Do not rely only on handouts. You can take data/information from different, authentic sources (like books, website, etc.) BUT express/organize all the collected material in **YOUR OWN WORDS**. Only then you will get good marks.

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Objective(s) of this Assignment:

To apply the concepts of sufficiency, completeness, UMVUE, and Bayesian estimation to solve statistical inference problems.

Question # 1

Marks:10

Let $X_1, X_2, \dots, X_n$ be a random sample from a Poisson distribution with parameter $\lambda > 0$.

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}, \quad x = 0, 1, 2, \dots$$

a): Show that $T = \sum_{i=1}^n X_i$ is a sufficient statistics λ .

4 Marks

b): Show that T is complete.

3 Marks

c): Find the UMVUE of λ .

3 Marks

Solution:

We are given a random sample $X_1, X_2, \dots, X_n$ from a Poisson distribution with parameter $\lambda > 0$:

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}, \quad x = 0, 1, 2, \dots$$

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(a) Show that $T = \sum_{i=1}^n X_i$ is a sufficient statistic for λ

Solution:

To establish sufficiency, we apply the **Factorization Theorem**.

The probability mass function of a Poisson random variable is

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}, x = 0, 1, 2, \dots$$

For a random sample $X_1, X_2, \dots, X_n$, the likelihood function is

$$L(\lambda; x_1, \dots, x_n) = \prod_{i=1}^n \frac{e^{-\lambda} \lambda^{x_i}}{x_i!}$$

This can be rewritten as

$$L(\lambda; x_1, \dots, x_n) = e^{-n\lambda} \lambda^{\sum_{i=1}^n x_i} \prod_{i=1}^n \frac{1}{x_i!}$$

Let

$$T = \sum_{i=1}^n X_i$$

Then the likelihood factors into:

- a term $e^{-n\lambda} \lambda^T$, which depends on the data only through T and on λ ,

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- and a term $\prod_{i=1}^n \frac{1}{x_i!}$, which does not depend on λ .

By the Factorization Theorem, $T = \sum_{i=1}^n X_i$ is a sufficient statistic for λ .

(b) Show that T is complete

Solution:

A statistic T is said to be **complete** if

$$E_{\lambda}[g(T)] = 0 \text{ for all } \lambda \Rightarrow g(T) = 0 \text{ almost surely.}$$

Since $X_1, \dots, X_n$ are independent $\text{Poisson}(\lambda)$ random variables, their sum

$$T = \sum_{i=1}^n X_i$$

has a Poisson distribution with parameter $n\lambda$:

$$T \sim \text{Poisson}(n\lambda).$$

The probability mass function of T is

$$P(T = t) = \frac{e^{-n\lambda} (n\lambda)^t}{t!}, t = 0, 1, 2, \dots$$

It is a known result that the Poisson family forms a **complete family of distributions**. Hence, T is a complete statistic.

(c) Find the UMVUE of λ

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Solution:

The sample mean is

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i = \frac{T}{n}.$$

Taking expectation,

$$E[\bar{X}] = \frac{E[T]}{n} = \frac{n\lambda}{n} = \lambda.$$

Thus, $\bar{X}$ is an unbiased estimator of λ and is a function of the sufficient statistic T .

Since T is both **complete and sufficient**, the **Lehmann–Scheffé theorem** guarantees that any unbiased estimator based on T is the **unique UMVUE**.

Therefore, the UMVUE of λ is

$$\hat{\lambda}_{\text{UMVUE}} = \frac{T}{n} = \bar{X}.$$

Question # 2

Marks:10

Let $X_1, X_2, \dots, X_n$ be a random sample from a normal distribution.

$$X_i \sim N(\theta, \sigma^2), \quad \sigma^2 \text{ known}$$

Assume the prior distribution of θ is:

$$\theta \sim N(\mu_0, \tau^2)$$

a): Find the Bayesian estimator of θ under squared error loss.

6Marks

b): State whether the Bayes estimator is unbiased or biased.

4Marks

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Solution:

We have a normal sample:

$$X_1, \dots, X_n \sim N(\theta, \sigma^2), \sigma^2 \text{ known}$$

Prior distribution of θ :

$$\theta \sim N(\mu_0, \tau^2)$$

(a) Find the Bayes estimator of θ under squared error loss

Solution:

Under squared error loss, the Bayes estimator is given by the **posterior expectation** of the parameter.

The likelihood function for the observed sample is

$$L(\theta) \propto \exp \left[-\frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \theta)^2 \right].$$

Assume a normal prior distribution for θ :

$$\pi(\theta) \propto \exp \left[-\frac{1}{2\tau^2} (\theta - \mu_0)^2 \right].$$

Combining the likelihood and the prior, the posterior distribution is

$$\pi(\theta | X_1, \dots, X_n) \propto \exp \left[-\frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \theta)^2 - \frac{1}{2\tau^2} (\theta - \mu_0)^2 \right].$$

By completing the square in θ , the posterior distribution is obtained as

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$$\theta | X \sim N \left[\frac{\frac{n\bar{X}}{\sigma^2} + \frac{\mu_0}{\tau^2}}{\frac{n}{\sigma^2} + \frac{1}{\tau^2}}, \frac{1}{\frac{n}{\sigma^2} + \frac{1}{\tau^2}} \right]$$

Hence, the Bayes estimator under squared error loss, which is the posterior mean, is

$$\hat{\theta}_{\text{Bayes}} = \frac{\frac{n\bar{X}}{\sigma^2} + \frac{\mu_0}{\tau^2}}{\frac{n}{\sigma^2} + \frac{1}{\tau^2}}$$

(b) State whether the Bayes estimator is unbiased or biased

Solution:

The expectation of the Bayes estimator is

$$E[\hat{\theta}_{\text{Bayes}}] = E \left[\frac{\frac{n\bar{X}}{\sigma^2} + \frac{\mu_0}{\tau^2}}{\frac{n}{\sigma^2} + \frac{1}{\tau^2}} \right] = \frac{\frac{n\theta}{\sigma^2} + \frac{\mu_0}{\tau^2}}{\frac{n}{\sigma^2} + \frac{1}{\tau^2}}$$

In general, this expectation is not equal to θ . Therefore, the Bayes estimator does not satisfy the unbiasedness condition.

Only in the special case where $\mu_0 = \theta$ does the estimator become unbiased.

Hence, the Bayes estimator

$$\hat{\theta}_{\text{Bayes}} = \frac{\frac{n\bar{X}}{\sigma^2} + \frac{\mu_0}{\tau^2}}{\frac{n}{\sigma^2} + \frac{1}{\tau^2}}$$

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is biased in general.

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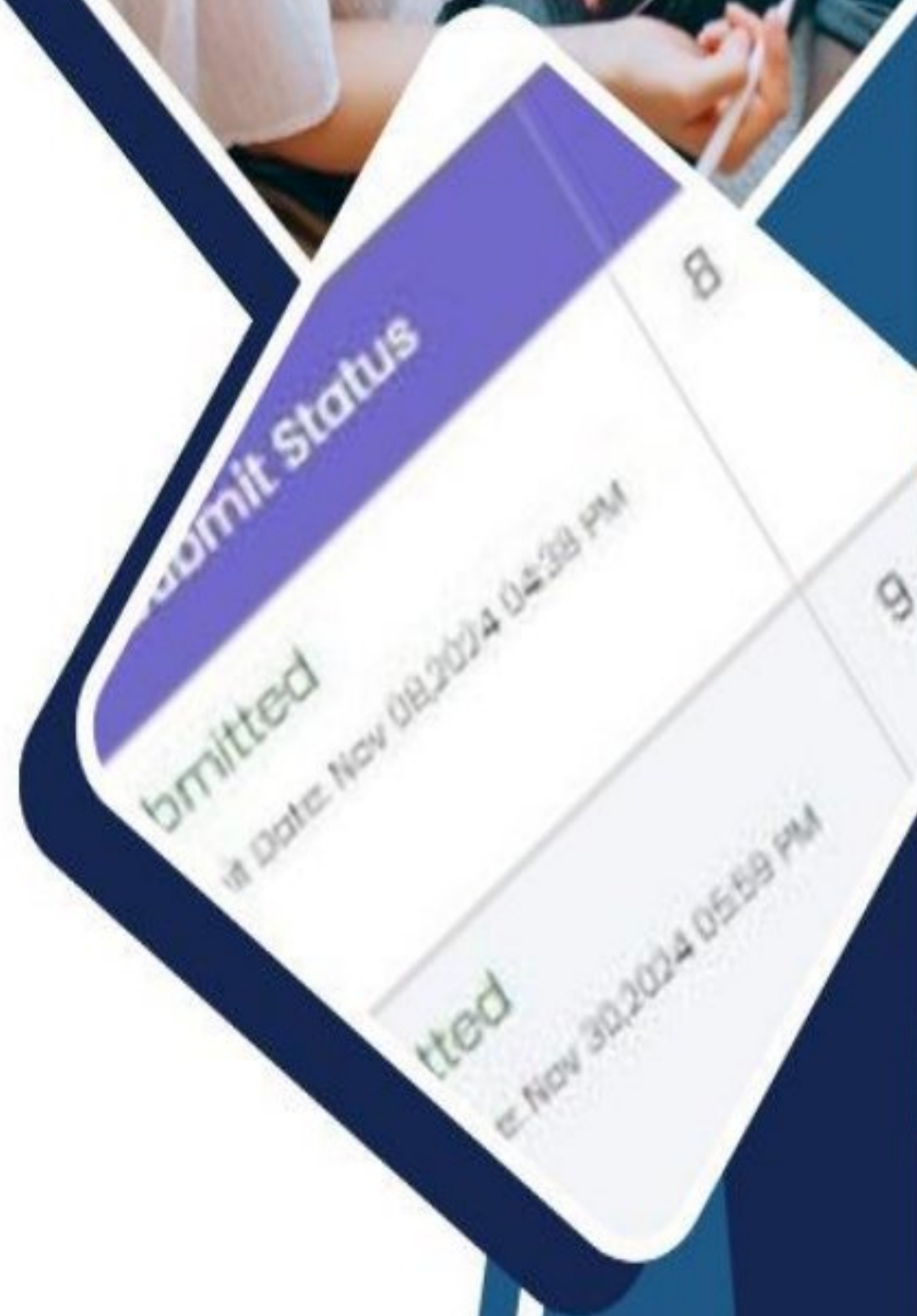
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